Generative Modelling with Stochastic Differential Equations

Exercises, week 5

Flows and guidance

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7 ODE and flow-based generative models

Setting for Exercises 7.1–7.3. Let q be a probability measure on $\mathbb{R}^n$ and $\bar{X}$ the forward OU process $\mathrm{d}\bar{X}_\tau = -\bar{X}_\tau \mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$, $\bar{X}_0 \sim q$, so that $\bar{X}_\tau \overset{d}{=} e^{-\tau}\bar{X}_0 + \sigma_\tau Z$ with $Z \sim \gamma_n = \mathcal{N}(0, \mathbb{I}_n)$ independent of $\bar{X}_0$ and $\sigma_\tau^2 = 1 - e^{-2\tau}$. Let $q_\tau = \mathrm{Law}(\bar{X}_\tau)$ and $s_\tau = \nabla \log q_\tau$ for $\tau > 0$; for $\tau > 0$, q_τ has a smooth positive density, solves the Fokker–Planck equation $\partial_\tau q_\tau = \nabla \cdot (x q_\tau) + \Delta q_\tau$, and satisfies Tweedie's formula $s_\tau(y) = \mathbb{E}[(e^{-\tau}\bar{X}_0 - y)/\sigma_\tau^2 \mid \bar{X}_\tau = y]$. Fix $T > 0$ and use the reverse time $t = T - \tau$. The *reverse probability flow ODE* (PF ODE) is

$$\frac{\mathrm{d}}{\mathrm{d}t} Y_t = Y_t + s_{T-t}(Y_t), \qquad Y_0 \sim q_T,$$

and $\mathrm{Law}(Y_t) = q_{T-t}$. On a grid $0 = t_0 < \cdots < t_K < T$ with $h_k = t_{k+1} - t_k$, put $\alpha_t := e^{-(T-t)}$, $\alpha_k := \alpha_{t_k}$, $\sigma_k := \sigma_{T-t_k}$, $\tilde{\beta}_k := (1 - e^{-2h_k})\sigma_{k+1}^2/\sigma_k^2$, and let $\hat{s}_{T-t_k}$ be a (learned) vector field and $\xi_k \overset{\text{i.i.d.}}{\sim} \gamma_n$. The DDPM, deterministic DDIM and stochastic DDIM updates are

$$\begin{aligned}
\text{(DDPM)} \quad & \hat{Y}_{k+1} = e^{h_k}\big[\hat{Y}_k + (1 - e^{-2h_k})\hat{s}_{T-t_k}(\hat{Y}_k)\big] + \sqrt{\tilde{\beta}_k}\,\xi_k, \\
\text{(DDIM)} \quad & \hat{Y}_{k+1} = e^{h_k}\hat{Y}_k + \big(e^{h_k}\sigma_k^2 - \sigma_k\sigma_{k+1}\big)\hat{s}_{T-t_k}(\hat{Y}_k), \\
\text{(DDIM-}\eta\text{)} \quad & \hat{Y}_{k+1} = e^{h_k}\hat{Y}_k + \Big(e^{h_k}\sigma_k^2 - \sigma_k\sqrt{\sigma_{k+1}^2 - \eta^2\tilde{\beta}_k}\Big)\hat{s}_{T-t_k}(\hat{Y}_k) + \eta\sqrt{\tilde{\beta}_k}\,\xi_k, \quad \eta \in [0,1].
\end{aligned}$$

Exercise 7.1 (The η-family preserves marginals)**.** (a) Show that $\Delta p = \nabla \cdot (p \nabla \log p)$ for every smooth positive density p.

(b) For $\eta \in [0,1]$, let Y^η solve $\mathrm{d}Y_t^\eta = [Y_t^\eta + (1+\eta^2)s_{T-t}(Y_t^\eta)]\mathrm{d}t + \sqrt{2}\,\eta\,\mathrm{d}\tilde{B}_t$ with $Y_0^\eta \sim q_T$, $t \in [0, T)$. Show that $p_t := q_{T-t}$ solves the Fokker–Planck equation of this SDE. Assuming uniqueness for this equation, conclude that $\mathrm{Law}(Y_t^\eta) = q_{T-t}$ for every η.

(c) Show that DDIM-η coincides with DDPM for $\eta = 1$ and with deterministic DDIM for $\eta = 0$.

(d) Show that for $\eta \neq \eta'$ the path laws of Y^η and $Y^{\eta'}$ on $[0, t]$, $t > 0$, are mutually singular.

Exercise 7.2 (DDPM from the OU bridge ($\star$))**.** Let $0 < \tau' < \tau$ and $h = \tau - \tau'$.

(a) Show that, conditionally on $\bar{X}_0 = x_0$ and $\bar{X}_\tau = y$, $\bar{X}_{\tau'} \sim \mathcal{N}(m(x_0, y), \tilde{\beta}\mathbb{I}_n)$ with

$$m(x_0, y) = \frac{e^{-\tau'}(1 - e^{-2h})}{\sigma_\tau^2} x_0 + \frac{e^{-h}\sigma_{\tau'}^2}{\sigma_\tau^2} y, \qquad \tilde{\beta} = (1 - e^{-2h})\frac{\sigma_{\tau'}^2}{\sigma_\tau^2}.$$

Hint: $\bar{X}_\tau = e^{-h}\bar{X}_{\tau'} + \sqrt{1 - e^{-2h}}\,Z'$ with $Z' \sim \gamma_n$ independent of $(\bar{X}_0, \bar{X}_{\tau'})$.

(b) Let $\hat{x}_0(y) := e^\tau(y + \sigma_\tau^2 \hat{s}(y))$. Show that $m(\hat{x}_0(y), y) = e^h[y + (1 - e^{-2h})\hat{s}(y)]$; with $\tau = T - t_k$, $\tau' = T - t_{k+1}$ this is the mean of the DDPM update.

(c) Show that if $\hat{s} = s_\tau$, then $\hat{x}_0(y) = \mathbb{E}[\bar{X}_0 \mid \bar{X}_\tau = y]$ and $m(\hat{x}_0(y), y) = \mathbb{E}[\bar{X}_{\tau'} \mid \bar{X}_\tau = y]$. Is the conditional law of $\bar{X}_{\tau'}$ given $\bar{X}_\tau = y$ Gaussian with variance $\tilde{\beta}\mathbb{I}_n$?

(d) Show that the three coefficients of DDPM (e^h, $e^h(1 - e^{-2h})$ and $\tilde{\beta}$) differ from those of Euler–Maruyama for the reverse SDE $\mathrm{d}\overleftarrow{X}_t = [\overleftarrow{X}_t + 2s_{T-t}(\overleftarrow{X}_t)]\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t$ (namely $1 + h$, $2h$ and $2h$) by $O(h^2(1 + 1/\tau))$ for $h \leq \tau/2 \leq 1$. Conclude that they agree to first order if and only if $h \ll \min\{1, \tau\}$.

Exercise 7.3 (DDIM for a Gaussian target ($\star$))**.** Let $q = \mathcal{N}(0, \varsigma^2 \mathbb{I}_n)$, $\varsigma \geq 0$, and use the exact score $\hat{s} = s$. Let $\lambda_t := \sigma_{T-t}/\alpha_t$ and $\tilde{Y}_t := Y_t/\alpha_t$. You may use that, along the PF ODE, $\frac{\mathrm{d}\tilde{Y}}{\mathrm{d}\lambda} = -\sigma_{T-t}\, s_{T-t}(\alpha_t \tilde{Y})$, and that DDIM is the Euler step of this equation in the variable λ.

(a) Compute q_τ and s_τ, and show that along the PF ODE $\frac{\mathrm{d}\tilde{Y}}{\mathrm{d}\lambda} = \frac{\lambda}{\varsigma^2+\lambda^2}\tilde{Y}$. Solve it.

(b) Show that one DDIM step multiplies $\tilde{Y}$ by $1 + \frac{\lambda_k(\lambda_{k+1}-\lambda_k)}{\varsigma^2+\lambda_k^2}$, whereas the exact flow multiplies it by $\sqrt{(\varsigma^2+\lambda_{k+1}^2)/(\varsigma^2+\lambda_k^2)}$. When are they equal?

(c) Let $\varsigma = 1$ and write $\alpha_k = \cos\theta_k$, $\sigma_k = \sin\theta_k$ with $\theta_k \in (0, \pi/2)$. Show that the exact flow is the identity, while one DDIM step multiplies $\hat{Y}_k$ by $\cos(\theta_k - \theta_{k+1})$. For a grid with $\theta_k - \theta_{k+1} = \Theta/K$ for all k, compute the total contraction and its behaviour as $K \to \infty$.

Exercise 7.4 (Instantaneous change of variables for linear fields)**.** Let $v_t(x) = A_t x + c_t$ with A, c continuous on $[0,1]$, let Φ_t be the flow of $\dot{Y}_t = v_t(Y_t)$, and let p_t be the density of $Y_t = \Phi_t(Y_0)$, $Y_0 \sim p_0$ (a positive C^1 density).

(a) Show that $\Phi_t(x) = M_t x + m_t$ with $\dot{M}_t = A_t M_t$, $M_0 = \mathbb{I}_n$, and $\dot{m}_t = A_t m_t + c_t$, $m_0 = 0$.

(b) Show Jacobi's formula $\frac{\mathrm{d}}{\mathrm{d}t}\log\det M_t = \operatorname{tr} A_t$, and deduce from the change-of-variables formula $p_t(\Phi_t(x)) = p_0(x)/\det M_t$ that $\frac{\mathrm{d}}{\mathrm{d}t}\log p_t(Y_t) = -\nabla \cdot v_t(Y_t)$.

(c) For $p_0 = \gamma_n$, $c = 0$ and a constant symmetric A, identify p_t and verify (b) by direct computation.

Setting for Exercises 7.5–7.6. Time $t \in [0,1]$ runs from noise ($t = 0$) to data ($t = 1$). Given a random path $(X_t)_{t\in[0,1]}$, C^1 in t, with $p_t := \mathrm{Law}(X_t)$, the *marginal velocity* is $v_t(x) := \mathbb{E}[\dot{X}_t \mid X_t = x]$. If the ODE $\dot{Y}_t = v_t(Y_t)$ is well posed, then $Y_0 \sim p_0$ implies $Y_t \sim p_t$ for all t.

Exercise 7.5 (Rectified-flow velocity via the score ($\star$))**.** Let $X_0 \sim \gamma_n$ and $X_1 \sim p_1$ be independent, $X_t = (1-t)X_0 + tX_1$, $p_t = \mathrm{Law}(X_t)$, and $g_u := p_1 * \mathcal{N}(0, u\mathbb{I}_n)$ for $u > 0$.

(a) Show directly that $v_t(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x] = (x - \mathbb{E}[X_0 \mid X_t = x])/t$ and, by Tweedie's formula, that $v_t(x) = \frac{x+(1-t)\nabla \log p_t(x)}{t}$ for $t \in (0,1)$.

(b) Let Y solve $\dot{Y}_t = v_t(Y_t)$ and set $W := Y_t/t$, $\lambda := (1-t)/t$. Show that $p_t(y) = t^{-n} g_{\lambda^2}(y/t)$ and that $\frac{\mathrm{d}W}{\mathrm{d}\lambda} = -\lambda \nabla \log g_{\lambda^2}(W)$.

(c) Let Y now solve the reverse PF ODE $\dot{Y}_t = Y_t + \nabla \log q_{T-t}(Y_t)$ for the OU process started from $q = p_1$, and set $W := Y_t/\alpha_t$ with $\alpha_t := e^{-(T-t)}$, and $\lambda := \sqrt{e^{2(T-t)} - 1}$. Show that W solves the same ODE as in (b).

(d) Conclude.

Exercise 7.6 (Reflow in one dimension ($\star$))**.** For a coupling (X_0, X_1) of p_0 and p_1, let $X_t = (1-t)X_0 + tX_1$, $v_t(x) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$, and let Z solve $\dot{Z}_t = v_t(Z_t)$, $Z_0 = X_0$; the *rectified coupling* is $\mathrm{Rectify}(X_0, X_1) := (Z_0, Z_1)$, a coupling of p_0 and p_1. Let $n = 1$, and assume that all the ODEs below have unique solutions for every initial condition.

(a) Show that $Z_1 = F(Z_0)$ for a strictly increasing function F.

(b) Let F be strictly increasing and consider the coupling $(Z_0, F(Z_0))$. Show that its rectified flow follows the straight lines $t \mapsto (1-t)Z_0 + tF(Z_0)$, so that $\mathrm{Rectify}(Z_0, F(Z_0)) = (Z_0, F(Z_0))$ and its straightness $\int_0^1 \mathbb{E}|(Z_1 - Z_0) - \dot{Z}_t|^2 \mathrm{d}t$ is zero. Conclude that one reflow reaches a fixed point.

(c) Assume that p_0, p_1 have continuous, strictly increasing distribution functions F_0, F_1 on $\mathbb{R}$, and that F in (a) is continuous. Show that $F = F_1^{-1} \circ F_0$ (the monotone, or quantile, coupling).

(d) For $p_0 = \mathcal{N}(0,1)$ and $p_1 = \mathcal{N}(m, \varsigma^2)$, $\varsigma > 0$, compute F and the velocity field of the second rectification.

8 Guidance and the Doob h-transform

Setting and notation for this chapter. *Guidance.* $\bar{X}$ is the forward Ornstein–Uhlenbeck (OU) process $\mathrm{d}\bar{X}_\tau = -\bar{X}_\tau \mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau$ on $\mathbb{R}^n$, $\bar{X}_0 \sim q_0$, with marginals $q_\tau = \mathrm{Law}(\bar{X}_\tau)$, $\sigma_\tau^2 = 1 - e^{-2\tau}$ and transition kernel $p_{\tau|0}(\cdot \mid x_0) = \mathcal{N}(e^{-\tau}x_0, \sigma_\tau^2 \mathbb{I}_n)$. A label $\mathbf{C}$ is attached to $\bar{X}_0$ and is independent of the driving Brownian motion B given $\bar{X}_0$ (more precisely, $(\bar{X}_0, \mathbf{C})$ is independent of B). We write $p_\tau(\mathbf{c} \mid x) = \mathbb{P}(\mathbf{C} = \mathbf{c} \mid \bar{X}_\tau = x)$ for the noisy classifier (a conditional density if $\mathbf{C}$ is continuous), $q_\tau(\cdot \mid \mathbf{c})$ for the law of $\bar{X}_\tau$ given $\mathbf{C} = \mathbf{c}$, and, for a guidance scale $\gamma > 0$,

$$s_\tau^\gamma(x) = \nabla \log q_\tau(x) + \gamma \nabla \log p_\tau(\mathbf{c} \mid x)$$

for the guided score. The (guided) reverse SDE runs in reverse time $t = T - \tau$: $\mathrm{d}\overleftarrow{X}_t = [\overleftarrow{X}_t + 2s_{T-t}^\gamma(\overleftarrow{X}_t)]\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t$; for $\gamma = 1$ and $\overleftarrow{X}_0 \sim q_T(\cdot \mid \mathbf{c})$ it has the law of $(\bar{X}_{T-t})_{t \le T}$ given $\mathbf{C} = \mathbf{c}$.

Doob and Feynman–Kac. A baseline diffusion $\mathrm{d}Y_t = b_t(Y_t)\mathrm{d}t + g_t(Y_t)\mathrm{d}B_t$, $0 \le t \le T$, has path law $\mathbb{Q}$, diffusion matrix $a_t = g_t g_t^\top$, generator $\mathcal{L}_t f = \langle b_t, \nabla f \rangle + \frac{1}{2}\operatorname{tr}(a_t \nabla^2 f)$, transition kernels $P_{s,t}$, and ν_t denotes the density of Y_t. A positive function $h_t(x)$ is *space-time harmonic* if $(\partial_t + \mathcal{L}_t)h = 0$; its *Doob drift* is $a_t \nabla \log h_t$. Given a terminal weight $G \ge 0$ with $0 < \mathbb{E}_\mathbb{Q}[G(Y_T)] < \infty$, the *Doob transform* is the path law $\mathbb{Q}^G$ with $\mathrm{d}\mathbb{Q}^G/\mathrm{d}\mathbb{Q} = G(Y_T)/\mathbb{E}_\mathbb{Q}[G(Y_T)]$, and its h-function is $h_t(x) = \mathbb{E}_\mathbb{Q}[G(Y_T) \mid Y_t = x]$. All functions are assumed smooth and integrable enough for the computations to make sense.

Exercise 8.1 (Output of linear-Gaussian guidance)**.** Let $\bar{X}_0 \sim \gamma_n$ and $\mathbf{C} = \bar{X}_0 + \eta\xi$ with $\xi \sim \gamma_n$ independent of $(\bar{X}_0, B)$ and $\eta > 0$. Fix $c \in \mathbb{R}^n$ and $\gamma > 1/2$, and write $u = e^{-2\tau}$ and $w_\tau = \sigma_\tau^2 + \eta^2 = 1 + \eta^2 - u$. In this model $q_\tau = \gamma_n$ for all τ and the guided score is $s_\tau^\gamma(x) = -x + \gamma e^{-\tau}(c - e^{-\tau}x)/w_\tau$.

(a) Run the guided reverse SDE $\mathrm{d}\overleftarrow{X}_t = [\overleftarrow{X}_t + 2s_{T-t}^\gamma(\overleftarrow{X}_t)]\mathrm{d}t + \sqrt{2}\,\mathrm{d}\tilde{B}_t$ from $\overleftarrow{X}_0 \sim \gamma_n$. Show that $\overleftarrow{X}_t$ is Gaussian with mean $\mu_\tau \in \mathbb{R}^n$ and covariance $P_\tau \mathbb{I}_n$, where, as functions of the forward level $\tau = T - t$,

$$\frac{\mathrm{d}\mu_\tau}{\mathrm{d}\tau} = \Big(1 + \frac{2\gamma u}{w_\tau}\Big)\mu_\tau - \frac{2\gamma e^{-\tau} c}{w_\tau}, \qquad \frac{\mathrm{d}P_\tau}{\mathrm{d}\tau} = 2\Big(1 + \frac{2\gamma u}{w_\tau}\Big)P_\tau - 2,$$

with $\mu_T = 0$ and $P_T = 1$. (Hint: $\mathrm{d}w_\tau/\mathrm{d}\tau = 2u$, so $e^\tau w_\tau^\gamma$ and $e^{2\tau} w_\tau^{2\gamma}$ are integrating factors.)

(b) Solve these equations and show that, with $r = \eta^2/(1+\eta^2)$, as $T \to \infty$,

$$\mathrm{Law}(\overleftarrow{X}_T) \;\longrightarrow\; \mathcal{N}\Big(c\,(1 - r^\gamma),\; \frac{\eta^2\,(1 - r^{2\gamma-1})}{2\gamma - 1}\,\mathbb{I}_n\Big).$$

(c) Check that this is $q_0(\cdot \mid c)$ for $\gamma = 1$. For $\gamma = 2$ and $\eta = 1$, compare it with the sharpened target $\pi_0^2 = \mathcal{N}(\frac{2}{3}c, \frac{1}{3}\mathbb{I}_n)$.

Exercise 8.2 (Classifier-free guidance for a Gaussian mixture)**.** Let $\mathbf{C}$ take values in a finite set $\mathcal{C}$ with $\mathbb{P}(\mathbf{C} = \mathbf{c}) = \pi_\mathbf{c} > 0$, and let $\bar{X}_0 \mid \mathbf{C} = \mathbf{c} \sim \mathcal{N}(\mu_\mathbf{c}, \mathbb{I}_n)$.

(a) Show that $q_\tau(\cdot \mid \mathbf{c}) = \mathcal{N}(e^{-\tau}\mu_\mathbf{c}, \mathbb{I}_n)$, that $p_\tau(\cdot \mid x)$ is the softmax of $\ell_\mathbf{c}(x) = \log \pi_\mathbf{c} + e^{-\tau}\langle \mu_\mathbf{c}, x \rangle - \frac{1}{2}e^{-2\tau}|\mu_\mathbf{c}|^2$, and

$$\nabla \log q_\tau(x \mid \mathbf{c}) = -x + e^{-\tau}\mu_\mathbf{c}, \qquad \nabla \log q_\tau(x) = -x + e^{-\tau}\bar{\mu}_\tau(x),$$

where

$$\bar{\mu}_\tau(x) := \sum_{\mathbf{c}'} p_\tau(\mathbf{c}' \mid x)\mu_{\mathbf{c}'}.$$

(b) With exact scores, compute the classifier-free guidance (CFG) score $\tilde{s}^\gamma_\tau(x) = \nabla \log q_\tau(x) + \gamma[\nabla \log q_\tau(x \mid \mathbf{c}) - \nabla \log q_\tau(x)]$ and check that it equals $s^\gamma_\tau = \nabla \log q_\tau + \gamma \nabla \log p_\tau(\mathbf{c} \mid \cdot)$. Ho and Salimans write CFG as $(1+w)\nabla \log q_\tau(x \mid \mathbf{c}) - w\nabla \log q_\tau(x)$; express $\tilde{s}^\gamma_\tau$ in this form and relate w to γ.

(c) Two symmetric classes: $\mathcal{C} = \{+, -\}$, $\pi_\pm = \frac{1}{2}$, $\mu_\pm = \pm\mu$. Show that $p_\tau(+ \mid x) = (1 + e^{-2\theta})^{-1}$ with $\theta = e^{-\tau}\langle \mu, x\rangle$, and that the CFG score for $\mathbf{c} = +$ minus the exact conditional score is

$$(\gamma - 1)\, e^{-\tau}(1 - \tanh\theta)\, \mu.$$

Interpret: for $\gamma > 1$ guidance adds a push in the direction μ, of size at most $2(\gamma-1)e^{-\tau}|\mu|$, strongest on the wrong side of the decision boundary and vanishing deep inside class $+$.

(d) In the setting of (c), show that $p_\tau(+ \mid x)^\gamma / (p_\tau(+ \mid x)^\gamma + p_\tau(- \mid x)^\gamma) = (1 + e^{-2\gamma\theta})^{-1}$: the powered classifier is the original one at temperature $1/\gamma$.

Exercise 8.3 (Brownian bridge as an h-transform)**.** Let $\mathrm{d}Y_t = \mathrm{d}B_t$ on $\mathbb{R}^n$, $Y_0 = x_0$, and fix $y \in \mathbb{R}^n$ and $T > 0$. Let φ_v be the density of $\mathcal{N}(0, v\mathbb{I}_n)$.

(a) For $\varepsilon > 0$ let $G_\varepsilon(x) = \varphi_\varepsilon(y - x)$. Compute $h^\varepsilon_t(x) = \mathbb{E}[G_\varepsilon(Y_T) \mid Y_t = x]$ and the Doob drift, and let $\varepsilon \downarrow 0$.

(b) Check directly that $h_t(x) = \varphi_{T-t}(y - x)$ satisfies $(\partial_t + \frac{1}{2}\Delta)h = 0$ on $[0, T) \times \mathbb{R}^n$.

(c) Show that $Y_t = x_0 + (y - x_0)\frac{t}{T} + (T - t)\int_0^t \frac{\mathrm{d}B_s}{T-s}$ solves $\mathrm{d}Y_t = \frac{y - Y_t}{T-t}\mathrm{d}t + \mathrm{d}B_t$ on $[0, T)$, is Gaussian with mean $x_0 + (y - x_0)t/T$ and covariance $\frac{t(T-t)}{T}\mathbb{I}_n$, and converges to y in L^2 as $t \uparrow T$.

(d) Show that these are the mean and covariance of B_t given $B_T = y$ (with $B_0 = x_0$), and that the time-t marginal of the h-transform is $\varphi_t(x - x_0)h_t(x)/h_0(x_0)$, the law of B_t given $B_T = y$.

Exercise 8.4 (Feynman–Kac kernels ($\star$))**.** Let $h > 0$ satisfy $(\partial_t + \mathcal{L}_t - V_t)h_t = 0$ for a continuous function V bounded below, and define, for $s \le t$,

$$P^{h,V}_{s,t} f(x) = \frac{1}{h_s(x)} \mathbb{E}_{\mathbb{Q}}\Big[e^{-\int_s^t V_r(Y_r)\mathrm{d}r} h_t(Y_t) f(Y_t) \;\Big|\; Y_s = x\Big].$$

Show that $P^{h,V}_{s,t}\mathbb{1} = \mathbb{1}$ and that $P^{h,V}$ satisfies the Chapman–Kolmogorov equations, so that it is a family of Markov kernels. (You may use the Feynman–Kac formula: if $(\partial_r + \mathcal{L}_r - V_r)h_r = 0$ on $[s, t]$, then $h_s(x) = \mathbb{E}_{\mathbb{Q}}[e^{-\int_s^t V_r(Y_r)\mathrm{d}r} h_t(Y_t) \mid Y_s = x]$.)

Exercise 8.5 (Powers of Feynman–Kac functions ($\star$))**.** Let $h > 0$ solve $(\partial_t + \mathcal{L}_t)h_t = V_t h_t$ for some function V, and let $\gamma > 0$.

(a) Show that h^γ solves the same equation with the potential $\gamma V + \frac{\gamma(\gamma-1)}{2}|g^\top \nabla \log h|^2$.

(b) Deduce what scale-γ guidance samples when the classifier term is itself an approximation $\hat{h}$ of the true h-function, with relative defect $V = (\partial_t + \mathcal{L}_t)\hat{h}/\hat{h}$.

Exercise 8.6 (Ignoring the label at initialisation)**.** Let $P^h_{0,T}$ be the transition kernel from time 0 to time T of the conditional reverse process, so that $q_0(\cdot \mid \mathbf{c}) = q_T(\cdot \mid \mathbf{c})P^h_{0,T}$. Suppose that a Markov process with the same kernels is started from q_T instead of $q_T(\cdot \mid \mathbf{c})$, so that the label is ignored at initialisation, and let ρ be its law at time T.

(a) Show that $\mathrm{KL}(q_0(\cdot \mid \mathbf{c}) \,\|\, \rho) \le \mathrm{KL}(q_T(\cdot \mid \mathbf{c}) \,\|\, q_T)$.

(b) Show that $\mathrm{KL}(q_T(\cdot \mid \mathbf{c}) \,\|\, q_T) \le e^{-2T}\mathcal{W}_2\big(q_0(\cdot \mid \mathbf{c}), q_0\big)^2/(2\sigma_T^2)$.

Exercise 8.7 (An x-independent defect only rescales)**.** (a) Let $h > 0$ satisfy $(\partial_t + \mathcal{L}_t)h_t = c(t)h_t$ for a continuous function c of time only. Show that $\tilde{h}_t = \exp\big(\int_t^T c(s)\mathrm{d}s\big)h_t$ is space-time harmonic, has the same drift correction $a\nabla \log \tilde{h} = a\nabla \log h$, and that the guided diffusion (drift $b + a\nabla \log h$, diffusion coefficient g) started from $h_0\nu_0/Z_0$ has marginals $\nu_t h_t/Z_t$ for all t, where $Z_t = \int \nu_t h_t$. (You may use Proposition 8.3.)

(b) Example: for $\mathrm{d}Y_t = \mathrm{d}B_t$ on $\mathbb{R}^n$, $Y_0 = 0$, and $h_t(x) = e^{\langle \theta, x\rangle}$ with $\theta \in \mathbb{R}^n$, compute the defect and the guided process, and verify directly that its marginals are proportional to $\nu_t h_t$.

9 Discrete diffusion models ($\star$)

Setting and notation for this sheet. Let $\Omega = \{0,1\}^n$. For $x \in \Omega$, $x^{[i]}$ is its i-th coordinate and $\varsigma_i(x)$ is x with coordinate i flipped; $\oplus$ is addition modulo 2 and $\pi = \mathrm{Unif}(\Omega)$.

- *Rate matrices.* A continuous-time Markov chain (CTMC) on a finite set is described by a rate matrix G_t: for $y \neq x$, $G_t(x,y) \geq 0$ is the rate of jumping from x to y, and $G_t(x,x) = -\sum_{y \neq x} G_t(x,y)$. Its transition probabilities satisfy $\mathbb{P}(X_{t+\epsilon} = y \mid X_t = x) = \mathbf{1}_{x=y} + \epsilon\, G_t(x,y) + o(\epsilon)$, and it acts on functions by $(G_t g)(x) = \sum_y G_t(x,y) g(y)$.

- *Forward chain.* Let $q = q_0$ be a law on Ω (the data), $\bar{X}_0 \sim q$, and let $(N_\tau^{[j]})_{\tau \geq 0}$, $j \leq n$, be independent Poisson processes of intensity $\frac{1}{2}$, independent of $\bar{X}_0$. Set $\bar{X}_\tau^{[j]} = \bar{X}_0^{[j]} \oplus N_\tau^{[j]}$. Its rate matrix is $\mathcal{L}(x, \varsigma_i(x)) = \frac{1}{2}$, $\mathcal{L}(x,x) = -\frac{n}{2}$ and $\mathcal{L}(x,y) = 0$ otherwise, i.e. $(\mathcal{L}g)(x) = \frac{1}{2}\sum_i (g(\varsigma_i(x)) - g(x))$. We write $q_\tau = \mathrm{Law}(\bar{X}_\tau)$, $K_{\tau',\tau}(x,y) = \mathbb{P}(\bar{X}_\tau = y \mid \bar{X}_{\tau'} = x)$ $K_\tau = K_{0,\tau}$ and $P_\tau g(x) = \sum_y K_\tau(x,y) g(y)$. In Exercise 9.1 you show that $K_\tau(x,y) = \prod_j \frac{1 \pm e^{-\tau}}{2}$, with $+$ if $y^{[j]} = x^{[j]}$ and $-$ otherwise; you may use this formula in the other exercises. In particular $q_\tau(y) = \sum_x q(x) K_\tau(x,y) > 0$ for all y when $\tau > 0$.

- *Time reversal.* Fix $T > 0$. The reversed process $\overleftarrow{X}_t = \bar{X}_{T-t}$, $t \in [0,T)$, is a CTMC with rate matrix $\overleftarrow{\mathcal{L}}_t(x,y) = \mathcal{L}(y,x)\, q_{T-t}(y)/q_{T-t}(x)$ for $y \neq x$. For the bit-flip chain, $\overleftarrow{\mathcal{L}}_t(x, \varsigma_i(x)) = \frac{1}{2} r_{T-t}^i(x)$, where

$$r_\tau^i(x) = \frac{q_\tau(\varsigma_i(x))}{q_\tau(x)}, \qquad \tau > 0,$$

  is the *discrete score*.

- *Learning.* Let $\Psi(a,b) = a\log(a/b) - a + b$ for $a, b > 0$, let $0 < \delta < T$, and let $\hat{r}_\tau^i(x) > 0$ be a model of $r_\tau^i(x)$ for $\tau \in [\delta, T]$. The *score entropy* is

$$\mathcal{J}_{\mathrm{SE}}(\hat{r}) = \int_\delta^T \mathbb{E}_{X \sim q_\tau} \sum_{i=1}^n \tfrac{1}{2}\, \Psi\big(r_\tau^i(X), \hat{r}_\tau^i(X)\big)\, \mathrm{d}\tau.$$

  The *learned reverse chain* $(\hat{Y}_t)_{t \in [0, T-\delta]}$ starts from $\hat{Y}_0 \sim \pi$ and has rates $\frac{1}{2}\hat{r}_{T-t}^i(x)$ from x to $\varsigma_i(x)$ (and 0 to non-neighbours).

Exercise 9.1 (Poisson parity and the forward kernel)**.** (a) Let $N \sim \mathrm{Poisson}(B)$. Show that $\mathbb{P}(N \text{ even}) = (1 + e^{-2B})/2$.

(b) Deduce that for $0 \leq \tau' \leq \tau$, $\mathbb{P}(\bar{X}_\tau^{[j]} = \bar{X}_{\tau'}^{[j]} \mid \bar{X}_{\tau'}) = (1 + e^{-(\tau - \tau')})/2$, that $K_{\tau',\tau}(x,y) = \prod_{j=1}^n \frac{1 \pm e^{-(\tau-\tau')}}{2}$ (with $+$ iff $y^{[j]} = x^{[j]}$), and that $(\bar{X}_\tau)$ is a Markov chain.

(c) Let k_τ be the 2×2 one-coordinate kernel. Show that $k_\tau = \frac{1}{2}J + e^{-\tau}(\mathbb{I}_2 - \frac{1}{2}J)$, where J is the all-ones 2×2 matrix, and deduce $K_{\tau'} K_\tau = K_{\tau' + \tau}$ and $K_\tau = e^{\tau\mathcal{L}}$, with $\mathcal{L}$ the rate matrix of the forward chain.

Exercise 9.2 (Spectral gap versus χ^2, and sharpness)**.** For $S \subseteq \{1, \dots, n\}$ let $\chi_S(x) = (-1)^{\sum_{i \in S} x^{[i]}}$. You may use that $\{\chi_S\}_S$ is an orthonormal basis of $L^2(\pi)$ with $\mathcal{L}\chi_S = -|S|\chi_S$, so that $P_\tau \chi_S = e^{-|S|\tau}\chi_S$, and that $\mathrm{KL}(q_T \| \pi) \leq n \log(1 + e^{-2T})$ for every q.

(a) Let $h_\tau = q_\tau/\pi$. Show $h_\tau = P_\tau h_0$ and $\chi^2(q_\tau\|\pi) \le e^{-2\tau}\chi^2(q\|\pi)$. Compute $\chi^2(\delta_x\|\pi)$ and explain why this bound is poor in high dimension.

(b) Show that for $q = \delta_x$ the bound on $\mathrm{KL}(q_T\|\pi)$ is sharp up to a factor 2 as $T \to \infty$.

Exercise 9.3 (Bounds on the discrete score)**.** Let $R^i_\tau(x, x_0) = K_\tau(x_0, \varsigma_i(x))/K_\tau(x_0, x)$ for $\tau > 0$. You may use that $\mathbb{E}[R^i_\tau(\bar X_\tau, \bar X_0) \mid \bar X_\tau = x] = r^i_\tau(x)$.

(a) Show that $R^i_\tau(x, x_0) = \tanh(\tau/2)$ if $x^{[i]} = x_0^{[i]}$ and $\coth(\tau/2)$ otherwise.

(b) Deduce that $\tanh(\tau/2) \le r^i_\tau(x) \le \coth(\tau/2)$ for every data law q, and that the total rate of the reverse chain at noise level τ is at most $\frac{n}{2}\coth(\tau/2) \le n(1 + 1/\tau)$.

(c) Show that for $n \ge 2$ the upper bound in (b) is attained at some state when q is a Dirac mass, and that it is not attained when $q = \pi$.

Exercise 9.4 (A single data point)**.** Let $q = \delta_{x^\star}$ for some $x^\star \in \Omega$. Call coordinate i of x *wrong* if $x^{[i]} \ne x^{\star[i]}$. You may use that $r^i_\tau(x) = \coth(\tau/2)$ if bit i of x is wrong and $r^i_\tau(x) = \tanh(\tau/2)$ if it is right.

(a) Show that the coordinates of $\overleftarrow{X}$ are independent two-state chains and that $w(t) := \mathbb{P}(\text{bit } i \text{ of } \overleftarrow{X}_t \text{ is wrong})$ solves $w' = -\frac{1}{2}\coth(\frac{T-t}{2})w + \frac{1}{2}\tanh(\frac{T-t}{2})(1-w)$, $w(0) = \frac{1-e^{-T}}{2}$. Check that $w(t) = \frac{1-e^{-(T-t)}}{2}$ and that $\overleftarrow{X}_t \to x^\star$ as $t \uparrow T$.

(b) Show that the expected number of flips of one coordinate of $\overleftarrow{X}$ on $[0, T]$ is $T/2$, as for the forward chain, even though the rates are unbounded. Why must this be so? What does the unbounded rate imply for uniformisation (Exercise 9.8), and what is the output error with early stopping at $T - \delta$?

Exercise 9.5 (Product data laws)**.** Let $q(x) = \prod_{j=1}^n p_j^{x^{[j]}}(1-p_j)^{1-x^{[j]}}$ with $p_j \in [0, 1]$.

(a) Show that q_τ is a product law with $p_j(\tau) = \frac{1}{2} + (p_j - \frac{1}{2})e^{-\tau}$.

(b) Compute $r^i_\tau(x)$ and show it depends on x only through $x^{[i]}$. Deduce that the reverse chain consists of n independent two-state chains, and that learning r decouples into n one-dimensional problems.

(c) Conversely, show that if for some $\tau > 0$ each $r^i_\tau(x)$ depends on x only through $x^{[i]}$, then q_τ and q are product laws. (So the dependence of $r^i_\tau(x)$ on the other coordinates is exactly what encodes the interactions.)

(d) Compute $\mathrm{KL}(q_T\|\pi)$ exactly and compare with the general bound $\mathrm{KL}(q_T\|\pi) \le ne^{-2T}$ of Exercise 9.2.

Exercise 9.6 (Masked diffusion on $\{0, 1, \#\}^n$)**.** Let $\Omega_\# = \{0, 1, \#\}^n$ and let q be a law on $\{0, 1\}^n \subset \Omega_\#$. In the forward chain, each coordinate that is not yet masked becomes $\#$ at rate 1, independently, and $\#$ is absorbing. Let $\alpha_\tau = e^{-\tau}$. For $y \in \Omega_\#$ let $U(y) = \{j : y^{[j]} \ne \#\}$ and let q_U be the marginal law of $x^{[U]}$ under q.

(a) Write the rate matrix and show that for $x \in \{0, 1\}^n$, $K_\tau(x, y) = \prod_j \big[\alpha_\tau \mathbf{1}_{y^{[j]} = x^{[j]}} + (1 - \alpha_\tau)\mathbf{1}_{y^{[j]} = \#}\big]$.

(b) Show $q_\tau(y) = \alpha_\tau^{|U|}(1-\alpha_\tau)^{n-|U|} q_U(y^{[U]})$ with $U = U(y)$.

(c) The time reversal of a finite-state CTMC with rate matrix $\mathcal{L}_\#$ and marginals q_τ has, at reverse time t and $\tau = T - t$, rates $\overleftarrow{\mathcal{L}}_t(y, y') = \mathcal{L}_\#(y', y)\, q_\tau(y')/q_\tau(y)$ for $y' \neq y$ with $q_\tau(y) > 0$. Using this, show that the reverse chain at reverse time t ($\tau = T - t$) only unmasks coordinates, and that for $i \notin U(y)$, $v \in \{0, 1\}$,

$$\overleftarrow{\mathcal{L}}_t(y, y^{i \leftarrow v}) = \frac{e^{-\tau}}{1 - e^{-\tau}}\, \mathbb{P}_{x \sim q}(x^{[i]} = v \mid x^{[U]} = y^{[U]}),$$

where $y^{i \leftarrow v}$ is y with coordinate i set to v.

(d) What must be learned? Compare with the bit-flip case.

(e) What replaces π? Show $\mathrm{TV}(q_T, \delta_{\#^n}) = 1 - (1 - e^{-T})^n \leq n e^{-T}$.

(f) Show that the true reverse chain reveals the coordinates in a uniformly random order (on the event that all are masked at time T) and draws each revealed value from $q(x^{[i]} = \cdot \mid \text{already revealed values})$.

Exercise 9.7 (Discrete Girsanov formula for a two-state chain)**.** Let $\Omega = \{0, 1\}$, $S > 0$, and let $a_t(x)$, $b_t(x)$ ($x \in \{0, 1\}$, $t \in [0, S]$) be continuous and positive. Let $\mathbb{Q}$ (resp. $\mathbb{P}$) be the law on $[0, S]$ of the CTMC that jumps from x to $1 - x$ at rate $a_t(x)$ (resp. $b_t(x)$), started from μ (resp. $\hat{\mu}$, with $\hat{\mu} > 0$). You may use the following fact: for a chain with continuous rates $\lambda_t(x)$, given $X_u = x$, the next jump time has survival function $\exp(-\int_u^s \lambda_r(x)\, \mathrm{d}r)$, $s \in [u, S]$.

(a) A path is determined by X_0, the number M of jumps and the jump times $U_1 < \cdots < U_M$. Show that under $\mathbb{Q}$ the triple (X_0, M, U) has density

$$\mu(x_0) \prod_{k=1}^{m} a_{u_k}(x_{k-1}) \exp\Big(-\int_0^S a_t(x(t))\, \mathrm{d}t\Big)$$

with respect to counting measure $\times$ Lebesgue measure on $\{u_1 < \cdots < u_m\}$, where $x_k = x_0 \oplus k$ and $x(t)$ is the path. Deduce $\mathbb{Q} \ll \mathbb{P}$ and a formula for $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$.

(b) Show that for every bounded continuous $f : [0, S] \times \Omega \to \mathbb{R}$,

$$\mathbb{E}_{\mathbb{Q}} \sum_{k=1}^{M} f(U_k, X_{U_k -}) = \mathbb{E}_{\mathbb{Q}} \int_0^S f(t, X_t)\, a_t(X_t)\, \mathrm{d}t.$$

Hint: use uniformisation (Exercise 9.8; you may assume it also holds for continuous rates): with $\Lambda = \sup_{t,x} a_t(x)$, the chain under $\mathbb{Q}$ can be realised by letting a Poisson process of intensity Λ propose candidate times s, at which the chain jumps with probability $a_s(X_{s-})/\Lambda$.

(c) Conclude that $\mathrm{KL}(\mathbb{Q} \| \mathbb{P}) = \mathrm{KL}(\mu \| \hat{\mu}) + \mathbb{E}_{\mathbb{Q}} \int_0^S \Psi(a_t(X_t), b_t(X_t))\, \mathrm{d}t$, where $\Psi(a, b) = a \log(a/b) - a + b$. (This is the discrete Girsanov formula for $\Omega = \{0, 1\}$.)

(d) Let $n = 1$, $S = T - \delta$, let $\mathbb{Q}$ be the law of the reversed bit-flip chain $(\overleftarrow{X}_t)_{t \leq T - \delta}$ and $\mathbb{P}$ that of the learned reverse chain $(\hat{Y}_t)_{t \leq T - \delta}$ (see the setting at the start of this sheet), with $\tau \mapsto \hat{r}_\tau$ continuous. Show that $\mathrm{KL}(\mathbb{Q} \| \mathbb{P}) = \mathrm{KL}(q_T \| \pi) + \mathcal{J}_{\mathrm{SE}}(\hat{r})$.

Exercise 9.8 (Uniformisation)**.** (a) Let G be a time-independent rate matrix on a finite set and $\bar{\lambda} \geq \max_x(-G(x, x))$. Show that $\widetilde{P} = \mathbb{I} + G/\bar{\lambda}$ is a stochastic matrix and that $e^{tG} = \sum_{k \geq 0} e^{-\bar{\lambda} t} \frac{(\bar{\lambda} t)^k}{k!} \widetilde{P}^k$. Deduce that if (Z_k) is a Markov chain with kernel $\widetilde{P}$ and (M_t) an independent Poisson process of intensity $\bar{\lambda}$, then $Y_t = Z_{M_t}$ has the finite-dimensional distributions of the CTMC with rate matrix G.

(b) Extend to piecewise-constant $t \mapsto G_t$.

(c) Consider the learned reverse chain of the bit-flip model on $[0, T-\delta]$, and assume its total jump rate satisfies $\frac{1}{2}\sum_i \hat{r}^i_\tau(x) \le \Lambda(\tau) := Cn(1+1/\tau)$ for all x and $\tau \in [\delta, T]$. It is simulated by uniformisation on a partition $0 = t_0 < \cdots < t_N = T - \delta$: on $[t_k, t_{k+1}]$, candidate times form a Poisson process of intensity $\bar{\lambda}_k \ge \sup_{t\in[t_k,t_{k+1}]} \Lambda(T-t)$. Compare the expected number of candidate times on $[0, T-\delta]$ for (i) a single $\bar{\lambda} = \Lambda(\delta)$; (ii) the partition $T - t_k = T(1+c)^{-k}$ for $k < N$, $t_N = T - \delta$, with $N = \lceil \log(T/\delta)/\log(1+c) \rceil$ and $\bar{\lambda}_k = \Lambda(T - t_{k+1})$.